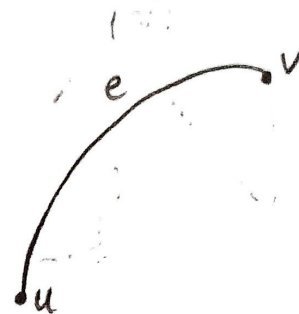


Graph: A triplet  $G = G(V, E, \phi)$  is called a graph if

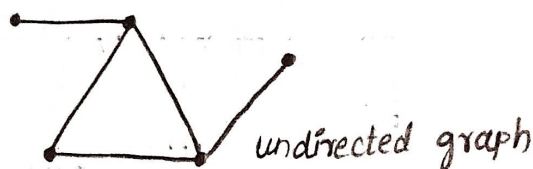
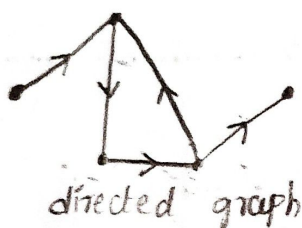
- i)  $V = V(G)$  is a vertex set
- ii)  $E = E(G)$  is an edge set
- iii) The map  $\phi: V \times V \rightarrow E$  by the rule

$$\phi((u, v)) = e \quad \forall (u, v) \in V \times V$$



Directed and undirected graphs

A graph  $G$  is called directed graph if each and every edge contains a direction. Otherwise, it is called undirected graph.

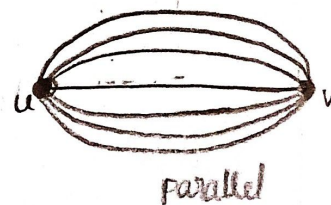


Loops and parallel edges (multiple)

An edge  $e$  is called a loop if its initial and final vertices are same.

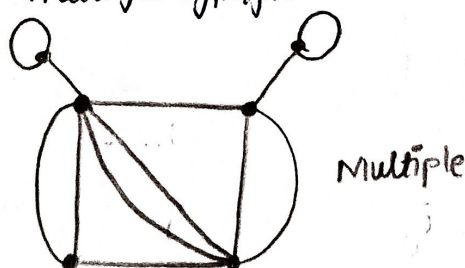
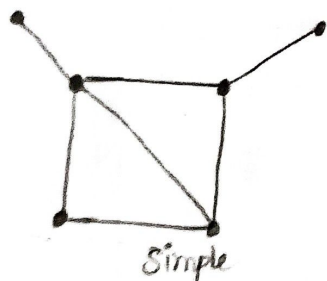


More than one edge is called parallel edge if its initial and final vertices are same.

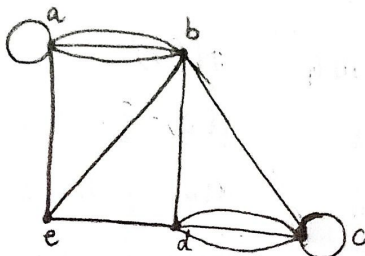
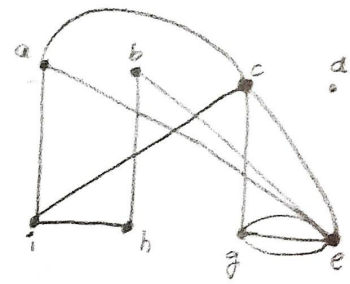
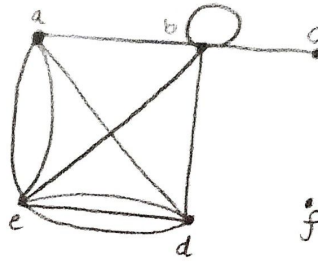
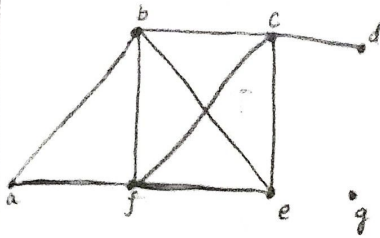


Simple and multiple graphs:

A graph  $G$  is called a simple if it has no loop and no parallel edges. Otherwise it is called multiple graph.



a) Describe degree and neighbourhood of a vertex in an undirected graph for the given graph.



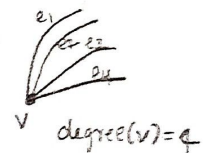
Find the degree of each vertex and neighbourhood of each vertex, and also identify isolated vertices and pendant vertices of these graphs.

### Degree of a vertex

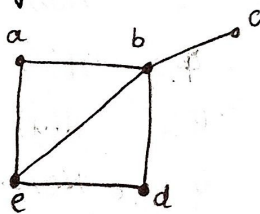
An edge  $e$  incident at the vertex  $v$  in  $G$  if the vertex  $v$  is either initial or final vertex

The degree of the vertex  $v$  in  $G$  is  $\text{degree}(v)$  and defined by

$$\text{deg}(v) = \text{the no. of edges incident at } v \text{ in } G$$



Example:

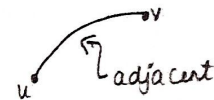


| vertex in $G$ | deg |
|---------------|-----|
| a             | 2   |
| b             | 4   |
| c             | 1   |
| d             | 2   |
| e             | 3   |

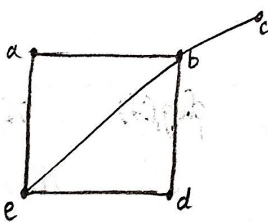
Neighbourhood of the vertex:

The neighbourhood of the vertex  $v$  in  $G$  is the set  $N(v)$  is defined by

$$N(v) = \{u \in V : u \text{ is adjacent to } v\}$$

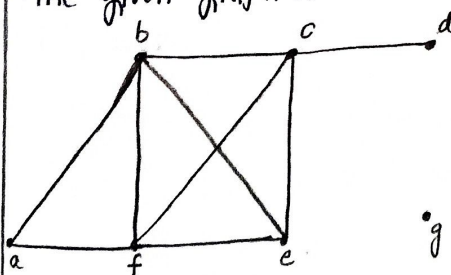


Ex:



| vertex in $G$ | $N(v)$                  |
|---------------|-------------------------|
| a             | $N(a) = \{b, e\}$       |
| b             | $N(b) = \{a, c, d, e\}$ |
| c             | $N(c) = \{b\}$          |
| d             | $N(d) = \{b, e\}$       |
| e             | $N(e) = \{a, b, d\}$    |

The given graph is



isolated vertices =  $\{g\}$  - no edge

pendant vertices =  $\{d\}$  - 1 edge

| vertex | deg |
|--------|-----|
| a      | 2   |
| b      | 4   |
| c      | 4   |
| d      | 1   |
| e      | 3   |
| f      | 4   |
| g      | 0   |

| vertex | $N(v)$                  |
|--------|-------------------------|
| a      | $N(a) = \{b, f\}$       |
| b      | $N(b) = \{a, c, e, f\}$ |
| c      | $N(c) = \{b, d, e, f\}$ |
| d      | $N(d) = \{c\}$          |
| e      | $N(e) = \{b, c, f\}$    |
| f      | $N(f) = \{a, b, c, e\}$ |
| g      | $N(g) = \{\}$           |



3) Explain the concepts of degree of a vertex and degree sequence of a graph. State necessary and sufficient conditions for a sequence to be graphic, and use these conditions to test the following sequences are graphic or not

i)  $5, 4, 3, 2, 1, 0$

ii)  $6, 5, 4, 3, 2, 1$

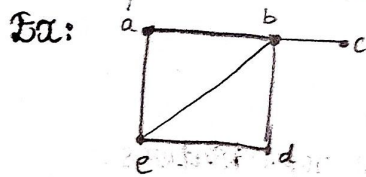
iii)  $2, 2, 2, 2, 2, 2$

iv)  $3, 3, 3, 2, 2, 2$  For each graphic sequence, construct one possible graph Degree of the vertex.

An edge  $e$  is an incident at the vertex  $v$  in  $G$  if the vertex  $v$  is either initial or final vertex

The degree of the vertex  $v$  in  $G$  is  $\deg(v)$  and defined by

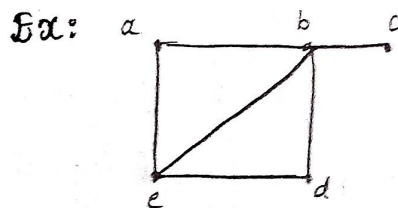
$\deg(v)$  = the no. of edges incident at  $v$  in  $G$



| vertex | $\deg(v)$ |
|--------|-----------|
| a      | 2         |
| b      | 4         |
| c      | 1         |
| d      | 2         |
| e      | 3         |

Degree sequence of the graph

Let  $\deg(v_i) = d_i$ ,  $1 \leq i \leq n$ , for any vertex  $v_i$  in  $G$ . then the degree sequence of  $G$  is  $\deg(G)$  and defined by  $\deg(G) = (d_1, d_2, \dots, d_n)$  where either  $d_1 \geq d_2 \geq \dots \geq d_n$  or  $d_1 \leq d_2 \leq \dots \leq d_n$



$$\deg(G) = (2, 4, 1, 2, 3)$$

$$= (4, 3, 2, 2, 1)$$

Graphic

A degree sequence  $\deg(G) = (d_1, d_2, d_3, \dots, d_n)$  is called a graphic if  $d_1 + d_2 + d_3 + \dots + d_n$  is even this is called necessary and sufficient condition for construction of a graph

i) The given sequence is

$$\deg(G) = (5, 4, 3, 2, 1, 0)$$

$$\therefore 5 + 4 + 3 + 2 + 1 + 0 = 15, \text{ not even}$$

$\Rightarrow \deg(G)$  is not graphic

$G$  is not constructed

ii) The given sequence is  $\deg(G) = (6, 5, 4, 3, 2, 1)$

$$\therefore 6 + 5 + 4 + 3 + 2 + 1 = 21, \text{ not even}$$

$\deg(G)$  is not graphic

$G$  is not constructed

iii) The given sequence is  $\deg(G) = (2, 2, 2, 2, 2, 2)$

$$\therefore 2+2+2+2+2+2 = 12, \text{ even}$$

$\Rightarrow \deg(G)$  is graphic

$\Rightarrow G$  is constructed



iv) The given sequence is

$$\deg(G) = (3, 3, 3, 2, 2, 2)$$

$$\therefore 3+3+3+2+2+2 = 15, \text{ not even}$$

$\deg(G)$  is not graphic

$G$  is not constructed.

4) Explain the following graph representations with examples.

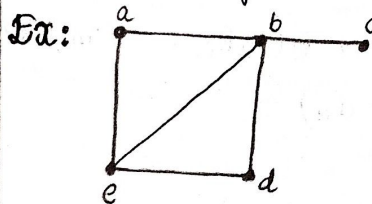
i) adjacency list

ii) adjacency matrix

iii) incidence matrix

For the given graph, construct all these three representations.

Adjacency list: The adjacent ~~list~~ of the vertex  $v$  in  $G$  is the collection of adjacency vertices of  $v$



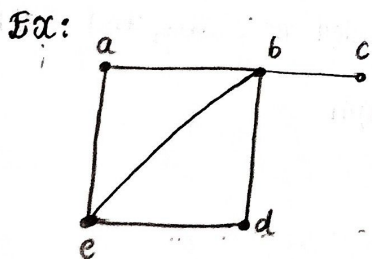
vertex in  $G$       list

|   |            |
|---|------------|
| a | b, e       |
| b | a, c, d, e |
| c | b          |
| d | b, e       |
| e | a, b, d    |

Adjacency matrix

The adjacency matrix of the graph  $G$  is denoted by  $A(G) = [a_{ij}]_{n \times n}$

Where  $a_{ij} = \begin{cases} 1 & \text{if } i \text{ is adjacent to } j \\ 0 & \text{otherwise} \end{cases}$



$\Rightarrow$

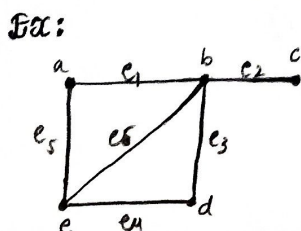
$A(G) =$

$$A(G) = \begin{bmatrix} a & b & c & d & e \\ a & 0 & 1 & 0 & 0 & 1 \\ b & 1 & 0 & 1 & 1 & 1 \\ c & 0 & 1 & 0 & 0 & 0 \\ d & 0 & 1 & 0 & 0 & 1 \\ e & 1 & 1 & 0 & 1 & 0 \end{bmatrix}_{5 \times 5}$$

Incidence matrix

The incidence matrix of the graph  $G$  is denoted by  $B(G) = [a_{ij}]_{m \times n}$

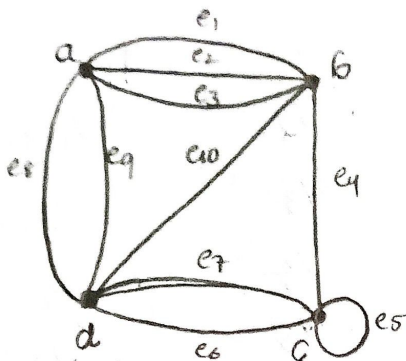
where  $a_{ij} = \begin{cases} 1 & \text{if the edge } i \text{ is incident at the vertex } j \\ 0 & \text{otherwise} \end{cases}$



$\Rightarrow B(G) =$

$$B(G) = \begin{bmatrix} a & b & c & d & e \\ e_1 & 1 & 1 & 0 & 0 & 0 \\ e_2 & 0 & 1 & 1 & 0 & 0 \\ e_3 & 0 & 1 & 0 & 1 & 0 \\ e_4 & 0 & 0 & 0 & 1 & 1 \\ e_5 & 1 & 0 & 0 & 0 & 1 \\ e_6 & 1 & 1 & 0 & 1 & 0 \end{bmatrix}$$

ii) The given graph is



| vertex in $G$ | list          |
|---------------|---------------|
| a             | a, b, b, d, d |
| b             | a, a, a, c, d |
| c             | c, c, d, d, b |
| d             | a, a, b, c, c |

Adjacency matrix

$$A(G) = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 0 & 3 & 0 & 2 \\ 3 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 2 & 1 & 2 & 0 \end{bmatrix} \end{matrix}_{4 \times 4}$$

Adj Incidence matrix

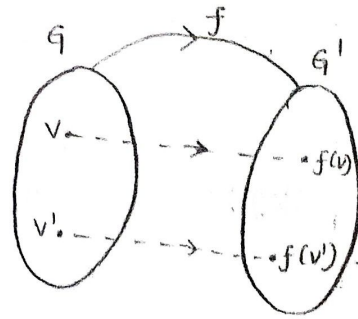
$$I(G) = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \\ e_6 \\ e_7 \\ e_8 \\ e_9 \\ e_{10} \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} \end{matrix}_{10 \times 4}$$



Graph isomorphism:

Two graphs  $G$  and  $G'$  are isomorphic, denoted by  $G \cong G'$  if there is a map  $f: V(G) \rightarrow V(G')$  such that  $f$  satisfies:

- i) one-one map
- ii) onto map
- iii)  $f$  preserves the adjacency



$$\text{i.e., } \{v, v'\} \in E(G) \Leftrightarrow \{f(v), f(v')\} \in E(G')$$

Necessary and sufficient conditions for Graph isomorphic

Step 1:  $|V(G)| = |V(G')|$  and  $|E(G)| = |E(G')|$

Step 2:  $\text{Deg}(G) = \text{Deg}(G')$

Step 3: The total no. of cycles of length  $k$  in  $G$   
 = the total no. of cycles of same length  $k$  in  $G'$

Step 4: Construct a bijective map (one-one and onto)  $f: V(G) \rightarrow V(G')$   
 such that  $f$  preserves adjacency

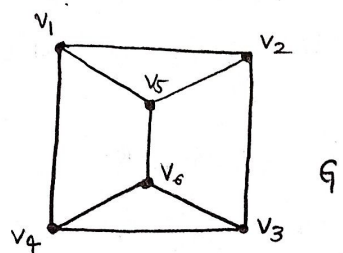
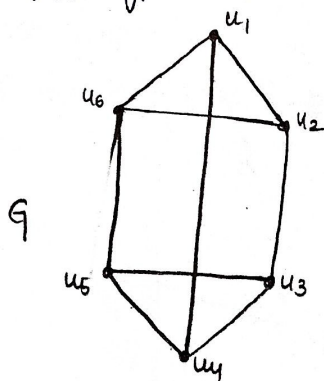
Steps: Applying this map  $f$  to show  $A(G) = A(G')$

Hence the given graphs are isomorphic,  $G \cong G'$

\*\*\*

8) State the necessary and sufficient conditions for two graphs to be isomorphic, and use these conditions to verify that the following graphs are isomorphic or not

i) The given two graphs are



Step 1:  $|V(G)| = 6 = |V(G')|$  and  $|E(G)| = 9 = |E(G')|$

Step 2:

| vertex in $G$ | $u_1$ | $u_2$ | $u_3$ | $u_4$ | $u_5$ | $u_6$ |
|---------------|-------|-------|-------|-------|-------|-------|
| deg           | 3     | 3     | 3     | 3     | 3     | 3     |

| vertex in $G'$ | $v_1$ | $v_2$ | $v_3$ | $v_4$ | $v_5$ | $v_6$ |
|----------------|-------|-------|-------|-------|-------|-------|
| deg            | 3     | 3     | 3     | 3     | 3     | 3     |

$$\text{deg}(G) = (3, 3, 3, 3, 3, 3) = \text{deg}(G')$$

Step 3: The no. of cycles of length 3, 4 & 6 in  $G =$   
 The no. of cycles of same length 3, 4 & 5 in  $G'$

Step 4: Define the bijective map  $f: V(G) \rightarrow V(G')$  by the rule:

$$\begin{aligned} f(u_1) &= v_5 & f(u_4) &= v_6 \\ f(u_2) &= v_2 & f(u_5) &= v_4 \\ f(u_3) &= v_3 & f(u_6) &= v_1 \end{aligned}$$

Step 5:

$$A(G) = \begin{matrix} & \begin{matrix} u_1 & u_2 & u_3 & u_4 & u_5 & u_6 \end{matrix} \\ \begin{matrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}_{6 \times 6}$$

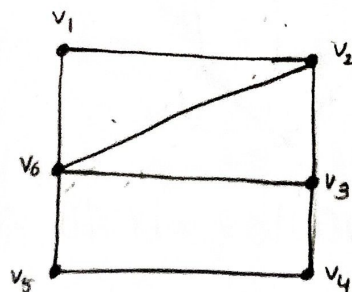
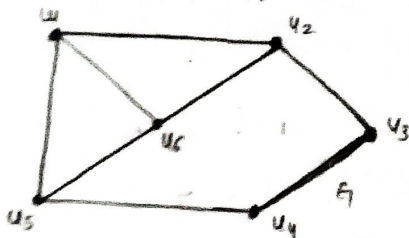
By applying  $f$ , to determine  $A(G') = f(A(G))$

$$A(G') = \begin{matrix} & \begin{matrix} v_5 & v_2 & v_3 & v_6 & v_4 & v_1 \end{matrix} \\ \begin{matrix} f(u_1) = v_5 \\ f(u_2) = v_2 \\ f(u_3) = v_3 \\ f(u_4) = v_6 \\ f(u_5) = v_4 \\ f(u_6) = v_1 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

Hence,  $A(G) = A(G')$  under  $f$

$\therefore$  The given two graphs are isomorphic  $G \cong G'$

(ii) The given two graphs are



Step 1:  $|V(G)| = 6 = |V(G')|$  and  $|E(G)| = 8 = |E(G')|$

|                       |       |       |       |       |       |       |
|-----------------------|-------|-------|-------|-------|-------|-------|
| Step 2: vertex in $G$ | $u_1$ | $u_2$ | $u_3$ | $u_4$ | $u_5$ | $u_6$ |
| deg                   | 3     | 3     | 2     | 2     | 3     | 3     |
| vertex in $G'$        | $v_1$ | $v_2$ | $v_3$ | $v_4$ | $v_5$ | $v_6$ |
| deg                   | 2     | 3     | 3     | 2     | 2     | 4     |

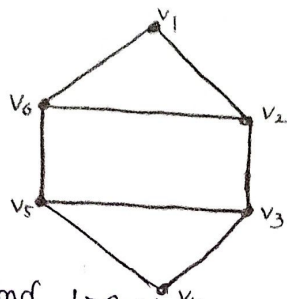
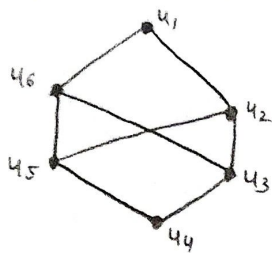
$$\deg(G) = (3, 3, 3, 3, 2, 2)$$

$$\deg(G') = (4, 3, 3, 2, 2, 2)$$

$$\deg(G) \neq \deg(G')$$

Hence  $G \not\cong G'$

iii) The given two graphs are



Step 1:  $|V(G)| = 6 = |V(G')|$  and  $|E(G)| = 8 = |E(G')|$

|                       |       |       |       |       |       |       |
|-----------------------|-------|-------|-------|-------|-------|-------|
| Step 2: vertex in $G$ | $u_1$ | $u_2$ | $u_3$ | $u_4$ | $u_5$ | $u_6$ |
| $\deg$                | 2     | 3     | 3     | 2     | 3     | 3     |
| vertex in $G'$        | $v_1$ | $v_2$ | $v_3$ | $v_4$ | $v_5$ | $v_6$ |
| $\deg$                | 2     | 3     | 3     | 2     | 3     | 3     |

$$\deg(G) = (3, 3, 3, 3, 2, 2)$$

$$\deg(G') = (3, 3, 3, 3, 2, 2)$$

$$\deg(G) = \deg(G')$$

Step 3: The no. of cycles of length 3 in  $G$  is 0, but the no. of cycles of length 3 in  $G'$  is 2

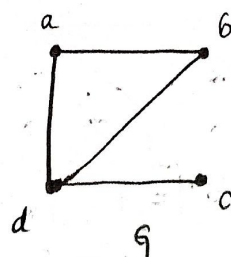
Hence  $G \not\cong G'$

\*\*\* 9) Two graphs are given by their adjacency matrices. use the conditions for the isomorphic graphs to check whether the graphs are isomorphic or not.

i) The give adjacent matrices for  $G$  and  $G'$  are

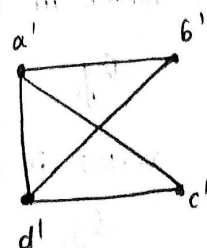
$$A(G) = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

$\Leftrightarrow$



$$A(G') = \begin{matrix} & \begin{matrix} a' & b' & c' & d' \end{matrix} \\ \begin{matrix} a' \\ b' \\ c' \\ d' \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

$\Leftrightarrow$





Step 1:  $|V(G)| = 4 = |V(G')|$

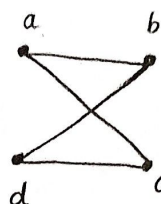
$|E(G)| = 4$  but  $|E(G')| = 5$

$|E(G)| \neq |E(G')|$

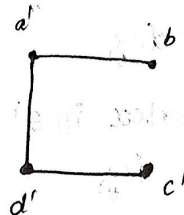
$G \neq G'$

ii) The given adjacent matrices for  $G$  and  $G'$

$A(G) = \begin{matrix} & a & b & c & d \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \end{matrix} \Leftrightarrow$



$A(G') = \begin{matrix} & a' & b' & c' & d' \\ \begin{matrix} a' \\ b' \\ c' \\ d' \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix} \Leftrightarrow$



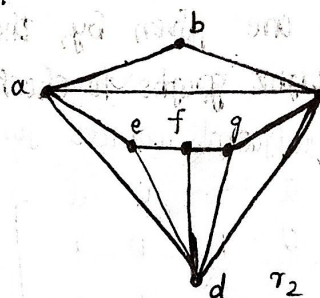
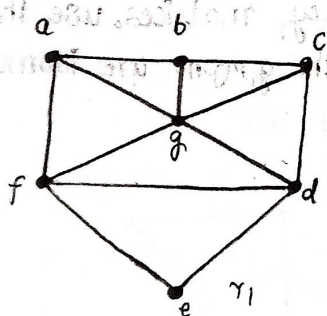
Step 1:  $|V(G)| = 4 = |V(G')|$  and  $|E(G)| = 4$  but  $|E(G')| = 3$

$|E(G)| \neq |E(G')|$

Hence  $G \neq G'$

7) Apply the necessary and sufficient conditions of isomorphic graphs to show that one and only one pair of these graphs is isomorphic.

a) consider the first two graphs.



Step 1:  $|V(T_1)| = 7 = |V(T_2)|$  and  $|E(T_1)| = 12 = |E(T_2)|$

Step 2: vertex in  $T_1$   $\begin{matrix} a & b & c & d & e & f & g \end{matrix}$

deg  $\begin{matrix} 3 & 3 & 3 & 4 & 2 & 4 & 5 \end{matrix}$

vertex in  $T_2$   $\begin{matrix} a' & b' & c' & d' & e' & f' & g' \end{matrix}$

deg  $\begin{matrix} 4 & 2 & 4 & 5 & 3 & 3 & 3 \end{matrix}$

$\deg(T_1) = (5, 4, 4, 3, 3, 3, 2)$  and  $\deg(T_2) = (5, 4, 4, 3, 3, 3, 2)$

$$\deg(\sigma_1) = \deg(\sigma_2)$$

step 3: The no. of cycles of length 3, 4, 5, 6 in  $G$  = The no. of cycles of, same length 3, 4, 5, 6 in  $G'$

step 4: Define the bijective map  $f: V(G_1) \rightarrow V(G_2)$  by the rule

$$\begin{array}{l|l|l} f(a) = e' & f(d) = a' & f(g) = d' \\ f(b) = f' & f(e) = b' & \\ f(c) = g' & f(f) = c' & \end{array} \quad 3, 5, 4, 2$$

step 5:

$$A(G) = \begin{array}{c|ccccccc} & a & b & c & d & e & f & g \\ \hline a & 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ b & 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ c & 0 & 1 & 0 & 1 & 0 & 0 & 1 \\ d & 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ e & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ f & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ g & 1 & 1 & 1 & 1 & 0 & 1 & 0 \end{array} \quad 7 \times 7$$

By the applying  $f$  we have

$$A(G_2) = \begin{array}{c|ccccccc} & e' & f' & g' & a' & b' & c' & d' \\ \hline f(a) = e' & 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ f(b) = f' & 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ f(c) = g' & 0 & 1 & 0 & 1 & 0 & 0 & 1 \\ f(d) = a' & 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ f(e) = b' & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ f(f) = c' & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ f(g) = d' & 1 & 1 & 1 & 1 & 0 & 1 & 0 \end{array}$$

$$\therefore A(G_1) = A(G_2)$$

and hence  $G_1 \cong G_2$